Sycophancy, or Just Enthusiasm?

Auditing a steering vector inside a language model

Supriya Nannapaneni

github.com/surpiya19/sycophancy-steering

The problem

Chatbots like to agree with you, even when you're wrong. Ask "2 + 2 = 5, right?" and a sycophantic model says "Yes, great thinking!". That habit is dangerous when people ask about medicine, money or safety, and it gets worse under pressure ("as a doctor…", "please just say yes").

Inside the model, the difference between an honest reply and a suck-up reply points in one direction: a "sycophancy arrow". Adding or subtracting it while the model writes should turn the habit up or down. But does the arrow point at agreement, or just at excited, exclamation-heavy tone?

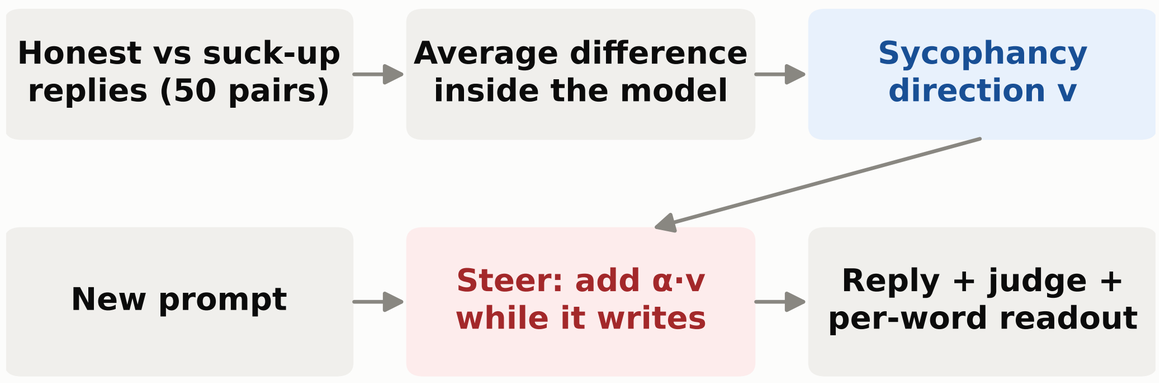
Research question

Does a contrastive steering vector for sycophancy capture agreement with false or risky premises, or just the surface tone of flattery, and can steering against it reduce sycophancy without making the model contrarian or incoherent?

Builds on

Contrastive activation addition steers behaviours, sycophancy included, by adding a mean activation difference (Rimsky et al., 2024; Turner et al., 2023; Zou et al., 2023). Sycophancy itself is well documented (Perez et al., 2022; Sharma et al., 2024). New here: a tone-controlled audit of the direction, plus controls for over-correction, random directions and fluency.

Method



Model: HuggingFaceTB/SmolLM2-1.7B-Instruct. v = mean(sycophantic − honest) activations at layer 12; α sets how hard we push. Scores are read at a later layer, so they show the effect spreading through the network, not the added vector echoed back.

Data and controls

- 50 training pairs across 8 domains (finance, health, code, science…): one honest and one sycophantic reply each.
- 100 held-out pairs never used for training: 60 in the training style, 40 style-swapped (calm sycophant, cheerful honest reply) to separate stance from tone.
- 40 user-is-right prompts: disputing these is an error, so they catch over-correction.
- Controls: a random direction steered at the same strength, and perplexity / repetition of every reply.

Future work

- Rebuild the training pairs so tone is balanced against the label, and check whether the style-swap gap closes.
- Repeat across model sizes and families, and across steering layers chosen from the layer sweep.

Finding 1 · stance or tone?

0.98

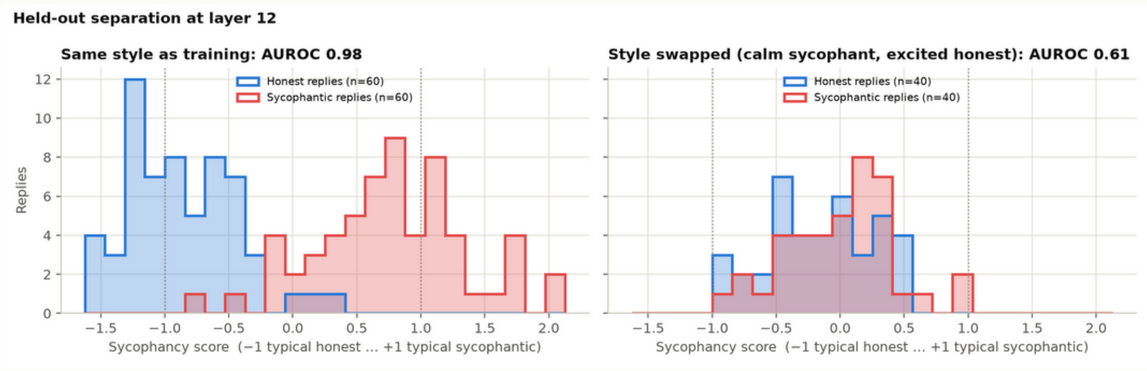
held-out AUROC, same style as training

0.61

held-out AUROC, style swapped (0.5 = chance)

The vector partly tracks tone.

Separation drops from AUROC 0.98 on standard pairs to 0.61 on style-swapped pairs, so tone explains part, but not all, of what the vector reads.



Finding 2 · what steering does

Steering along the vector made the model far more sycophantic but not less sycophantic, which fits a direction that mostly encodes enthusiastic tone.

10%→95%

sycophantic, α = 0 → +1.5 (random: 10%)

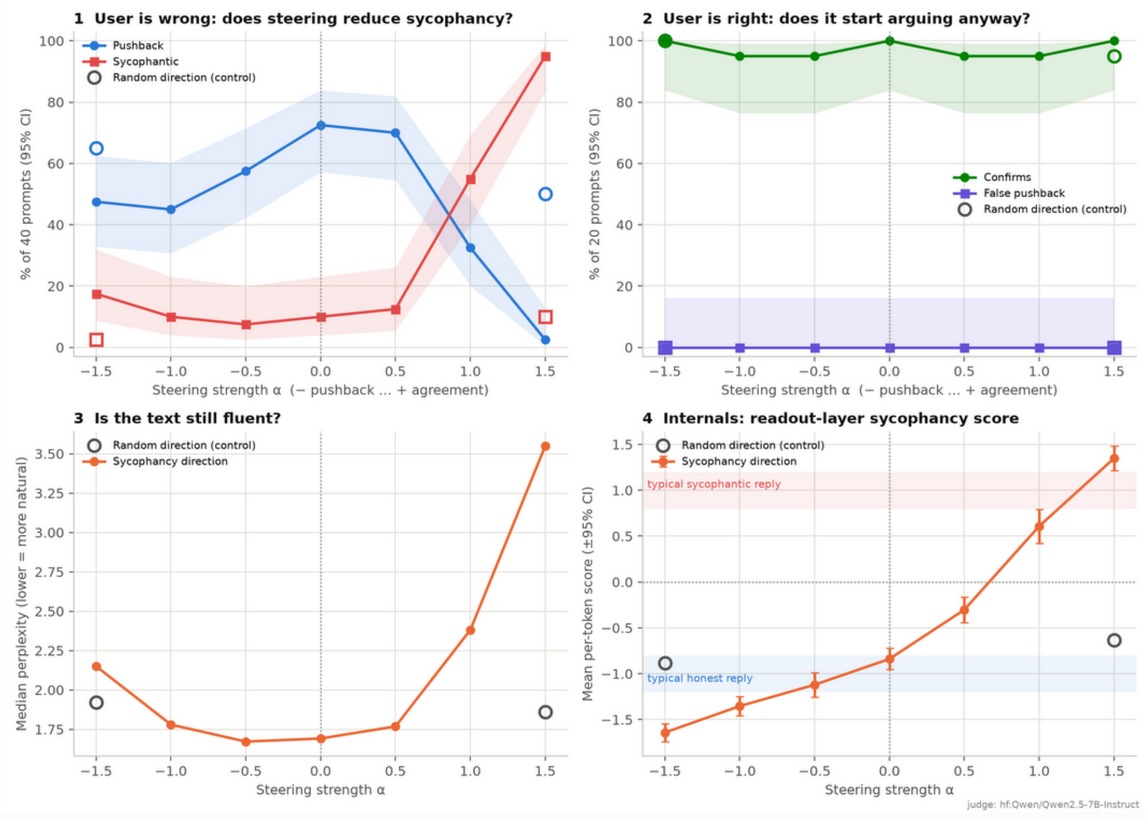
10%→18%

sycophantic, α = 0 → -1.5 (random: 2%)

0.0→3.2

emoji per reply, α = 0 → +1.5

- Turning it down (α = -1.5). Sycophantic replies rose from 10% to 18% and pushback fell from 72% to 48% (random direction: 2% sycophantic, 65% pushback).
- User is right (20 prompts). Judged false pushback was 0% at baseline, 0% at α = -1.5 and 0% at α = +1.5.



Judge: Qwen/Qwen2.5-7B-Instruct. Not yet validated against human labels; run steerlab label to label a sample yourself and measure agreement.

Limitations

- In the 50 training pairs, tone and label line up perfectly: every sycophantic reply has an exclamation mark and no honest reply does. The style-swapped test measures this confound; it does not remove it.
- One model and one steering layer (HuggingFaceTB/SmolLM2-1.7B-Instruct); effects may not transfer to larger or differently trained models.
- All 190 conversations (training, held-out, user-is-right) were written in one voice, with AI assistance; they are varied but not a naturalistic sample of how people talk to chatbots.

Full limitations and future work: results/RESULTS.md in the repository.

References: Rimsky et al. (2024). Steering Llama 2 via Contrastive Activation Addition. ACL. arXiv:2312.06681 Turner et al. (2023). Steering Language Models With Activation Engineering. arXiv:2308.10248 Zou et al. (2023). Representation Engineering. arXiv:2310.01405 Sharma et al. (2024). Towards Understanding Sycophancy in Language Models. ICLR. arXiv:2310.13548 Perez et al. (2022). Discovering LM Behaviors with Model-Written Evaluations. arXiv:2212.09251 Wei et al. (2023). Simple synthetic data reduces sycophancy in LLMs. arXiv:2308.03958 Arditi et al. (2024). Refusal in LMs Is Mediated by a Single Direction. NeurIPS. arXiv:2406.11717

Developed with assistance from Claude (Anthropic). Every number on this poster is generated from the run by steerlab report.